



MECH MAINT- BOILER Department
SIMHADRI



Ref: SMTTP/MECH MAINT-/2022-23/LMI/623630

Date:

24/11/2022(16:09:03)

Subject: Review of LMI “BOILER THERMAL EXPANSION INDICATORS AND THEIR USE ”

OPERATION & MAINTENANCE

MECHANICAL MAINTENANCE

(PRESSURE PARTS)

REF.: SMPP/O&M/MM/PP/

Date: 24.11.2022

Subject: Review of LMI “BOILER THERMAL EXPANSION INDICATORS AND THEIR USE ”

LMI No: LMI/OGN/OPS/MECH/006

The above approved LMI was revised in month of Aug '2021. The envisaged changes in LMI are incorporated as per details at revision index of LMI.

Put up for approval please.

Submitted by:

राहुल जैन - वरिष्ठ प्रबंधक

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On: 24/11/2022(16:09:03) **Ref:** SMTTP/MECH MAINT-/2022-23/LMI/623630

This LMI with doc ref no- LMI/OGN/OPS/MECH/006/ ISSUE-1; REV.-8.00; NOV '2022 has been reviewed and changes are incorporated as envisaged in OGN ref doc: COS-ISO-00-OGN/OPS/MECH/006.

May please accord approval for this revised version of LMI.

Submitted by:

सुरेश बाबु कोल्ला - उप महाप्रबंधक

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On:24/11/2022(16:49:05) **Ref:**SMTTP/MECH MAINT-/2022-23/LMI/623630

Following changes incorporated in revised LMI at page 3 of 10:

Additional frequency of readings added during start up to full load after modification and long shut down of unit.

May please be approved.

Submitted by:

ओम प्रकाश - अपर महाप्रबंधक

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On:25/11/2022(11:33:45) **Ref:**SMTTP/MECH MAINT-/2022-23/LMI/623630

Submitted for approval pls

Submitted by:

झिमलि धर - अपर महाप्रबंधक

Jhimli Dhar - 006460 (ADDL. GENERAL MANAGER ,EEMG)
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On:29/11/2022(09:33:34) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/623630

Revised LMI may please be accorded approval.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

Mathew Eipe Kovoov - 005637 (GENERAL MANAGER ,MAINTENANCE)
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On:29/11/2022(11:01:37) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/623630

कृपया अनुमोदन प्रदान किया जाए।

May be approved please.

Submitted by:

राजेश कुमार झा - महाप्रबंधक

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On:29/11/2022(11:46:30) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/623630

Approved & Submitted by:

गिरीश चन्द्र चौकसे - मुख्य महाप्रबंधक

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On:30/11/2022(12:29:20) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/623630

File Closed by:

Jhimli Dhar - 006460 (ADDL. GENERAL MANAGER ,EEMG) (JHIMLIDHAR@NTPC.CO.IN,
9650998085) ,Simhadri

On:30/11/2022(16:03:28)

NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/MECH/006/

ISSUE NO.1; REVISION NO. 8.00; DATE: NOV '2022

**BOILER THERMAL EXPANSION
INDICATORS AND THEIR USE**

APPROVED FOR

Implementation.....

HOP, SIMHADRI

Date-----

LMI APPROVAL DOCUMENTS

<u>1.0 TITLE OF LMI</u>	BOILER THERMAL EXPANSION INDICATORS
<u>2.a) DOCUMENT CODE</u>	<u>LMI/OGN/OPS/MECH/006/</u>
b) REVISION/ISSUE NO.&DATE	<u>ISSUE-1; REV.-8.00; NOV '2022</u>
<u>3.ORIGINAL/REF.DOCUMENT</u>	<u>COS-ISO-00-OGN/OPS/MECH/006</u>
<u>4.PREPARED BY:</u> Sr. Mgr. (MM-PP)	SIGNATURE-----
<u>5. CHECKED BY:</u> DGM(MM-BMD) AGM (BMD) GM (MTC.) GM (O&M)	SIGNATURE_____ SIGNATURE_____ SIGNATURE_____ SIGNATURE_____
<u>8. REVIEWED AND APPROVED BY</u>	SIGNATURE:----- (HOP, SIMHADRI)
<u>9.DATE OF ISSUE</u>	
<u>10.INDEX LIST</u>	ENCLOSED.
<u>11.DISTRIBUTION LIST</u>	ENCLOSED.

NTPC	LMI	SIMHADRI
TITLE: BOILER THERMAL EXPANSION INDICATORS REV: 8.00 DATE: NOV'2022		
REF NO. LMI/OGN/OPS/MECH/006/		PAGE : 1 OF 10
<u>1.0 INDEX</u>		
<u>SECTION</u>	<u>TITLE</u>	<u>PAGE</u>
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2.0	INTRODUCTION	2
3.0	SCOPE	2
4.0	REFERENCE DOCUMENT	2
5.0	SUPERSEDED DOCUMENT	2
6.0	SPECIFIC INSTRUCTIONS	3
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15.0	EXPANSION SURVEY RECORD	9-10

NTPC	LMI	SIMHADRI
TITLE: BOILER THERMAL EXPANSION INDICATORS REV: 8.00 DATE: NOV'2022		
REF NO. LMI/OGN/OPS/MECH/006/		PAGE : 2 OF 10
<u>2.0 INTRODUCTION</u> The two-pass drum type boiler installed at Simhadri is suspended from top and expands downwards as it is being brought up to pressure and temperature. The boiler is provided with thermal expansion indicators at different predetermined locations to measure, monitor and record the expansions at different hot conditions. If the boiler is not allowed to expand freely it may lead to failure of furnace tube attachments to backstay hot channel and buck stay sliding connections. This could lead to prolonged outages. The purpose of boiler expansion indicators is not for absolute accuracy but to see quickly and easily where any major problems in boiler movement or any inconsistent movements of one position or level, relative to another has occurred. The use of indicators at the upper level is primarily to ensure that the supporting system in the roof dead space is functioning correctly and not trying to twist or tilt the boiler and that the horizontal movements are not being affected by the internal piping. In this area, absolute errors can be small but significant on a percentage basis. The bottom level indicators are intended to show the effect of any major restrictions to vertical movements and any undue forces acting on the bottom headers from the feed piping. Indicators at intermediate levels are intended principally to quickly isolate the area where a restriction is occurring and to see the effects of restrictions occurring at other levels. Indicators on drum, down comers and headers are to ensure correct support, freedom from rotational stresses on stub welds and the correct return to datum. The objective of this LMI is to establish a system of periodic examination, measurement and recording of boiler expansion movements. <u>3.0 SCOPE</u> The scope of this LMI is to detail the locations and methods that are to be adopted for the periodic recording of expansion movements. It refers essentially to the measurement of boiler casing and steel movement. <u>4.0 REFERENCE DOCUMENT</u> Operation guidance note COS-ISO-00-OGN/OPS/MECH/006 circulated by operation services-NTPC corporate center. <u>5.0 SUPERSEDED DOCUMENT</u> BOILER THERMAL EXPANSION INDICATORS REV: 5.00 DATE: JAN'2020		

NTPC	LMI	SIMHADRI												
TITLE: BOILER THERMAL EXPANSION INDICATORS REV: 8.00														
DATE: NOV'2022														
REF NO. LMI/OGN/OPS/MECH/006/		PAGE : 3 OF 10												
<u>SPECIFIC INSTRUCTIONS</u>														
a) Integrity of expansion indicators at predetermined locations is to be ensured. b) Regular inspection to ensure free expansion of boiler. c) Measurement and recording of boiler expansion movements at different hot conditions. d) Exception recording and taking up with manufacturer/erector.														
<u>METHODOLOGY OF IMPLEMENTATION AND RESPONSIBILITY CENTERS</u>														
Integrity of expansion indicators to be checked during every overhaul by operation staff and necessary remedial action is to be taken to ensure presence and functional utility of static pointer and movable scale (attached to the boiler). Datum checking and correction, if required, before and after the boiler is filled, during light up and boiling out periods, acid cleaning time, during steam purging, during safety valve pressure setting, startup to full load after modifications and shutdown. Frequency : During overhaul Responsibility :HOS Mech. Maint (PP) b) Visual inspection of boiler at six monthly intervals and after every overhaul at all elevations to ensure that no rigid structure/beam/support is fouling with the pressure part components. Frequency : Every six months Responsibility : HOS Mech. Maint (PP) c) After every overhaul the boiler expansion movements are to be recorded during the first light after overhaul at the following Pressure/temperature intervals <table border="1"> <thead> <tr> <th>INTERVAL</th><th>DRUM PRESSURE(BAR)</th><th>SATURATION TEMP (DEG. CENTIGRADE)</th></tr> </thead> <tbody> <tr> <td>01</td><td>10</td><td>180</td></tr> <tr> <td>02</td><td>40</td><td>250</td></tr> <tr> <td>03</td><td>150</td><td>343</td></tr> </tbody> </table> Frequency : After overhaul Responsibility : HOS Mech. Maint (PP) d) If any ash accumulations are there same are to be recorded along with expansion readings of boiler horizontal duct positive, negative slope and FG duct expansions. Frequency: After overhaul ; Responsibility: HOS MM(RF) e) Exception recording and reporting to manufacturer- HOD (BMD)			INTERVAL	DRUM PRESSURE(BAR)	SATURATION TEMP (DEG. CENTIGRADE)	01	10	180	02	40	250	03	150	343
INTERVAL	DRUM PRESSURE(BAR)	SATURATION TEMP (DEG. CENTIGRADE)												
01	10	180												
02	40	250												
03	150	343												

8.0 EXCEPTION REPORTING

The exceptions encountered during inspection as compared to erection protocols are to be reported as per the following format and to be taken up with manufacturer HOD (BMD).

Description of exception	Unit No.	Location of exception	Remarks

9.0 RESPONSIBILITIES

The responsibility of carrying out different tasks in relation to inspection and maintenance of boiler expansion indicators as detailed in the LMI are with HOS (Pressure Parts)

10.0 RECORD KEEPING

The records to be maintained under this LMI include

- a) Compliance record of the activities detailed under this LMI
- b) Reports of expansion survey carried out
- c) Exception reporting

The above records are to be maintained by Area Engineer - Pr. Parts Group.

11.0 REVIEW

This document shall be reviewed by Head of the Project on Two-yearly basis.

NTPC		LMI		SIMHADRI
TITLE: BOILER THERMAL EXPANSION INDICATORS REV: 8.00				
DATE:NOV’2022				
REF NO. LMI/OGN/OPS/MECH/006/			PAGE : 5 OF 10	
12.0 REVISION INDEX (For changes/revision)				
Sl. No.	Rev. no.	Date	Details	Approved by
01	1.00	27.01.06	Cl. 730 (c) modified Compliance sheet modified.	
02	2.00	27.02.09	(a) Cl. 5 modified (b) Cl. 11 modified (c) Cl. 7.0 (c) modified	
03	3.00	13.10.2011	(a) Cl. 5.0 modified (b) Cl. 7.0 (a) & (c) modified (c) Cl. 11 modified (d) Cl. 9.0 modified	
04	004	30.04.13	i) Responsibility changed from SS (PP) to DGM (PP) at Responsibilities.	
			ii) Inclusion of Unit#3 & 4 in compliance record	
			iii) Page No. 9 added forchange in elevations	
			iv) Distribution List	
05	005	18.01.2020	i) Approval routing changed at “LMI APPROVAL DOCUMENT.”	
			ii) Review Frequency changed from "three years" to "TWO years"	
			iii) Distribution List	
			iv) X,Y,Z coordinate system legend includedin expansion survey record	
06	006	03.08.2020	i) In Clause 7, Methodology of implementation and responsibility centers, operation staff added andsub-clause d) added.	

NTPC		LMI		SIMHADRI	
TITLE: BOILER THERMAL EXPANSION INDICATORS REV: 8.00 DATE:NOV’2022					
REF NO. LMI/OGN/OPS/MECH/006/				PAGE : 6 OF 10	
Sl. No.	Rev. no.	Date	Details	Approved by	
07	007	09.08.2021	i) In clause -15, provision for Expansion indicator reading at EL-68m is mentioned in ST-1 format, whereas there is no provision for expansion measurement at that location in St-I. ii) At the same elevation indicator is available in ST-II units, hence the format in clause-15 is changed accordingly for St-II.		
08	008	24.11.2022	Changes done in bold as per COS-ISO-00-OGN/OPS/MECH/006 Rev 2, Oct 2022		

[illegible]

NTPC	14. COMPLIANCE RECORD	SIMHADRI
TITLE: BOILER THERMAL EXPANSION INDICATORS REV: 8.00		
DATE: NOV'2022		
REF NO. LMI/OGN/OPS/MECH/006/		PAGE : 8 OF 10
UNIT NO	ACTIVITY	COMPLIANCE
I/II/III/IV		

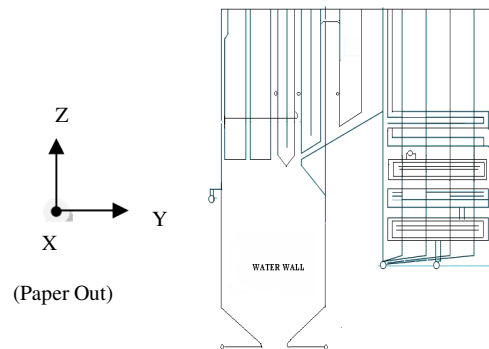
NTPC	15. EXPANSION SURVEY RECORD (STG - I)			SIMHADRI
TITLE: BOILER THERMAL EXPANSION INDICATORS REV: 8.00				
DATE: NOV'2022				
REF NO. LMI/OGN/OPS/MECH/006/			PAGE : 9 OF 10	
LOCATION	X-AXIS	Y AXIS	Z-AXIS	Remarks
Boiler drum El. 75m				
LHS				
RHS				
Eco-inlet EL 45m 2 nd pass				
Corner-1				
Corner-4				
Wind box Bottom El. 40m First pass				
Corner-1				
Corner-2				
Corner-3				
Corner-4				
Furnace guides El 23m First pass (Two-way movement only)				
Front	XXXXXXXX			
LHS		XXXXXXXX		
Rear	XXXXXXXX			
RHS		XXXXXXXX		
Ring header El. 12 m				
Corner-1				
Corner-2				
Corner-3				
Corner-4				

COORDINATES:

X: Boiler Left to Right

Y: Boiler Front to Rear

Z: Boiler Top to Bottom



NTPC	15. EXPANSION SURVEY RECORD (STG- II)			SIMHADRI
TITLE: BOILER THERMAL EXPANSION INDICATORS REV: 8.00 DATE: NOV’2022				
REF NO. LMI/OGN/OPS/MECH/006/			PAGE : 10 OF 10	
LOCATION	X-AXIS	Y AXIS	Z-AXIS	Remarks
Boiler drum El. 75m				
LHS				
RHS				
Furnace guides El 68m (First pass)				
LHS				
RHS				
2 nd pass bottom EL 43m				
Corner-1				
Corner-2				
Corner-3				
Corner-4				
Wind box Top El. 40 m				
Corner-1				
Corner-2				
Corner-3				
Corner-4				
Bottom Ring header El. 12 m				
Corner-1				
Corner-2				
Corner-3				
Corner-4				

COORDINATES:

X: Boiler Left to Right

Y: Boiler Front to Rear

Z: Boiler Top to Bottom

